The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 Jul 93.

INCH-POUND

MIL-S-19500/402B
8 April 1993
SUPERSEDING
MIL-S-19500/402A
14 December 1973

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPE 2N3739
JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

Types	P_T 1/ T_C = +25°C	P_T 2/ T_C = +100°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_{STG} and T_J	$R_{\theta JC}$
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N3739	20	10	325	300	6.0	0.5	1.0	-55 to +200	7.5

1/ Derate linearly, 0.114 W/°C for $T_C \geq +25°C$.
2/ Derate linearly, 0.100 W/°C for $T_C \geq +100°C$.

1.4 Primary electrical characteristics at T_A = 25°C.

Limit	h_{FE1} 1/ V_{CE} = 10 V dc, I_C = 10 mA dc	h_{FE3} 1/ V_{CE} = 10 V dc, I_C = 100 mA dc	V_{BE} V_{CE} = 10 V dc, I_C = 100 mA dc	$V_{CE(SAT)}$ I_C = 250 mA dc, I_B = 25 mA dc	C_{obo} V_{CB} = 100 V dc, I_E = 0 mA dc, 100 kHz≤f≤1 MHz	$\|h_{fe}\|$ V_{CE} = 10 V dc, I_C = 100 mA dc, f = 10 MHz	Switching t_{on}	Switching t_{off}
			V dc	V dc	pF		µs	µs
Min	30	40				1		
Max		200	1	2.5	20	6	1.5	3.5

1/ Pulsed (see 4.5.1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 12.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein.

3.4 Marking. Devices shall be marked in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and as specified herein.

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4.3 Screening (JANTX AND JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels only
9	Not applicable
11	h_{FE2} and I_{CBO1}
12	See 4.3.1
13	Subgroup 2 of table I herein, ΔI_{CBO1} = 100% of initial value or 10 μA dc, whichever is greater; Δh_{FE2} = ±20% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

2N3739 - - - - $V_{CB} \geq 50$ V dc, T_J = +162.5°C ±12.5°C

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX), of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

a. Subgroup 3: Condition for intermittent operation life are as follows:

$V_{CB} \geq 10$ V dc, ΔT_J between cycles ≥ 100°C, $t_{on} = t_{off} \geq 3$ min. for 2,000 cycles.
No heat sink or forced-air cooling on the devices shall be permitted.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

a. Subgroup 6: Condition for intermittent operation life (accelerated) are as follows:

$V_{CB} \geq 10$ V dc, ΔT_J between cycles ≥ 100°C, $t_{on} = t_{off} \geq 3$ min. for 6,000 cycles.
No heat sink or forced-air cooling on the devices shall be permitted.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 5.0 mA dc	$V_{(BR)CEO}$	300		V dc
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 325 V dc	I_{CBO1}		0.1	mA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{CE} = 325 V dc, V_{BE} = 1.5 V dc	I_{CEX}		0.5	mA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6.0 V dc	I_{EBO}		0.1	mA dc
Base to emitter, non-saturated voltage	3066	Test condition B, V_{CE} = 10 V dc, I_C = 100 mA dc; pulsed (see 4.5.1)	V_{BE}		1	V dc
Collector to emitter voltage (saturated)	3071	I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.75	V dc
Collector to emitter voltage (saturated)	3071	I_C = 250 mA dc; I_B = 25 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		2.5	V dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE1}	30		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE2}	30		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 100 mA dc; pulsed (see 4.5.1)	h_{FE3}	40	200	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 250 mA dc; pulsed (see 4.5.1)	h_{FE4}	25		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE5}	10		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 325 V dc	I_{CBO2}		1.0	mA dc
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 100 mA dc pulsed (see 4.5.1)	h_{FE6}	15		
Subgroup 4						
Pulse response:						
Turn-on time		V_{CC} = 150 V dc; I_C = 500 mA dc; I_B = 50 mA dc	t_{on}		1.5	µs
Turn-off time		V_{CC} = 150 V dc; I_C = 500 mA dc; I_{B1} = I_{B2} = 50 mA dc	t_{off}		3.5	µs
Magnitude of common emitter, small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 100 mA dc; f = 10 MHz	$\|h_{FE}\|$	1.0	6	
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 20 V dc; I_C = 100 mA dc;	h_{fe}	35	300	
Open circuit output capacitance	3236	V_{CB} = 100 V dc; I_E = 0 mA dc; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		20	pF
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; t = 1 s; 1 cycle; (see figure 2)				
Test 1		V_{CE} = 80 V dc; I_C = 250 mA dc				
Test 2		V_{CE} = 290 V dc; I_C = 6 mA dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area (switching)	3053	Load condition C (unclamped inductive load) (see fig. 3)				
		T_A = +25°C; duty cycle ≤ 10%; R_s = 1Ω; $t_r = t_f \leq 500$ ns				
Test 1		t_p approx 8 ms (Vary to obtain I_C); R_{BB1} = 100Ω; $V_{BB1} \geq 10$ V dc; $R_{BB2} = \infty$; V_{BB2} = 0 V dc; $V_{CC} \geq$ 100 V dc; I_C = 500 mA dc; The coil used shall provide a minimum inductance of 3.5 mH at 500 mA with max. dc resistance of 0.5 ohm (For reference only: Acme T58220, or equivalent)				
Test 2		t_p approx 8 ms (Vary to obtain I_C); R_{BB1} = 100Ω; $V_{BB1} \geq 10$ V dc; $R_{BB2} = \infty$; V_{BB2} = 0 V dc; $V_{CC} \geq$ 100 V dc; I_C = 100 mA dc; The coil used shall provide a minimum inductance of 25 mH at 100 mA with max. dc resistance of 1.0 ohm (For reference only: Triad C-48u, centertapped, or equivalent.)				
Safe operating area (switching)		T_A = +25°C; duty cycle ≤ 10%; t_p approx 8 ms (Vary to obtain I_C) $V_{CC} \geq$ 100 V dc; I_C = 1 A dc; R_s = 1Ω; clamp voltage = 300 V dc				
Electrical measurements		See table II, steps 1 and 2				

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups B and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 325 V dc	I_{CBO1}		0.1	mA dc
2	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 100 mA dc; pulsed (see 4.5.1)	h_{FE3}	40	200	
3	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 325 V dc	ΔI_{CBO}	100 percent or 10 μA dc, whichever is greater.		
4	Forward-current transfer ratio	3076	V_{CE} = 10 mA dc; I_C = 100 mA dc; pulsed (see 4.5.1)	Δh_{FE1}	±25 percent change from initial recorded value.		

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

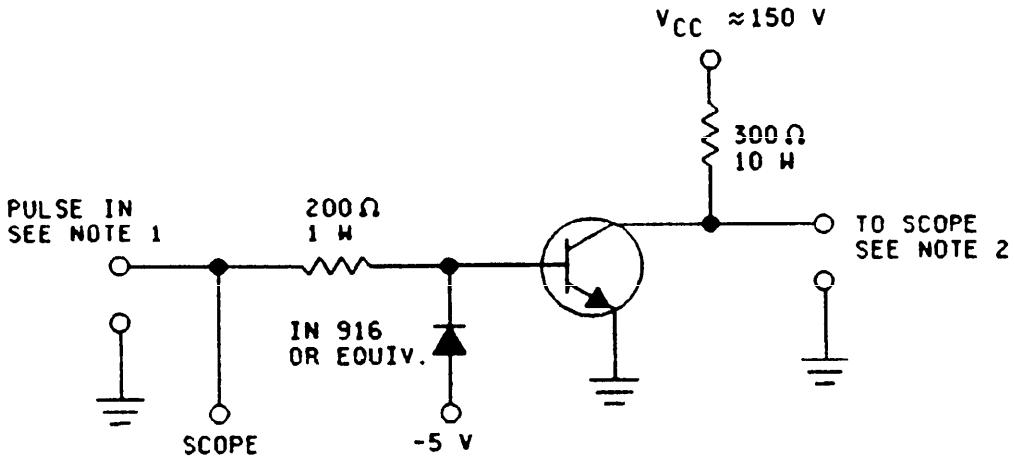
b. Subgroups 3 and 6, see table II herein, steps 3 and 4.

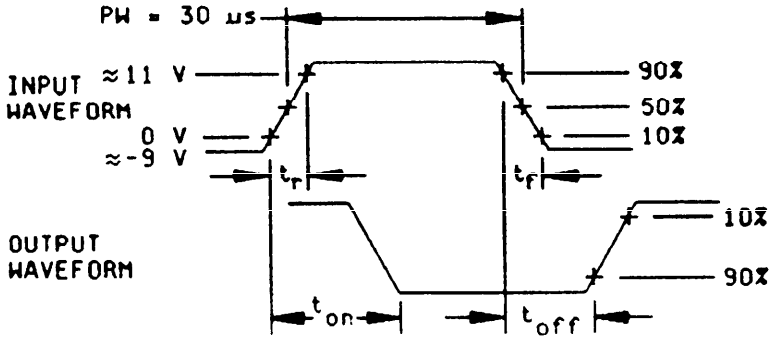
2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1 and 2.

b. Subgroup 6, see table II herein, steps 3 and 4.

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NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 20 ns; duty cycle ≤ 1 percent; generator source impedance shall be 50 ohms.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 50 pF; rise time ≤ 2.0 ns.

FIGURE 1. Pulse response test circuit.

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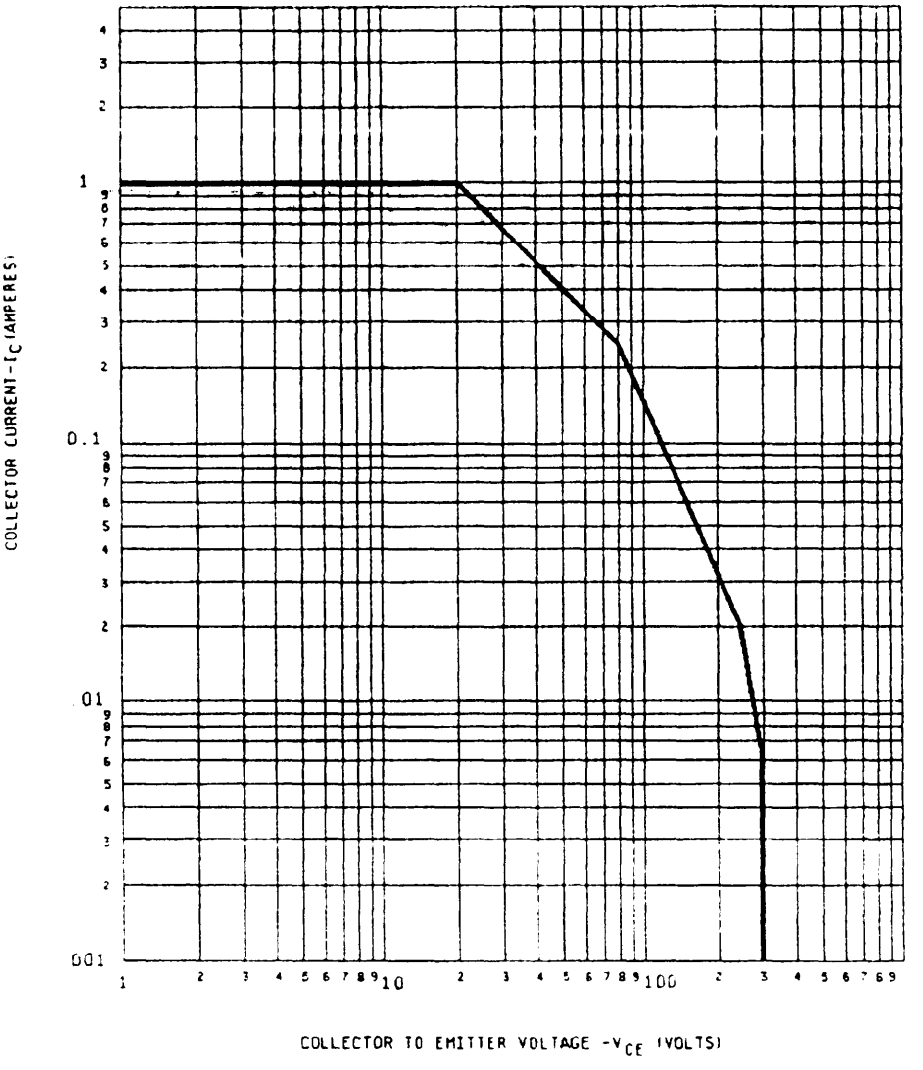


FIGURE 2. Maximum safe operating area graph (continuous dc).

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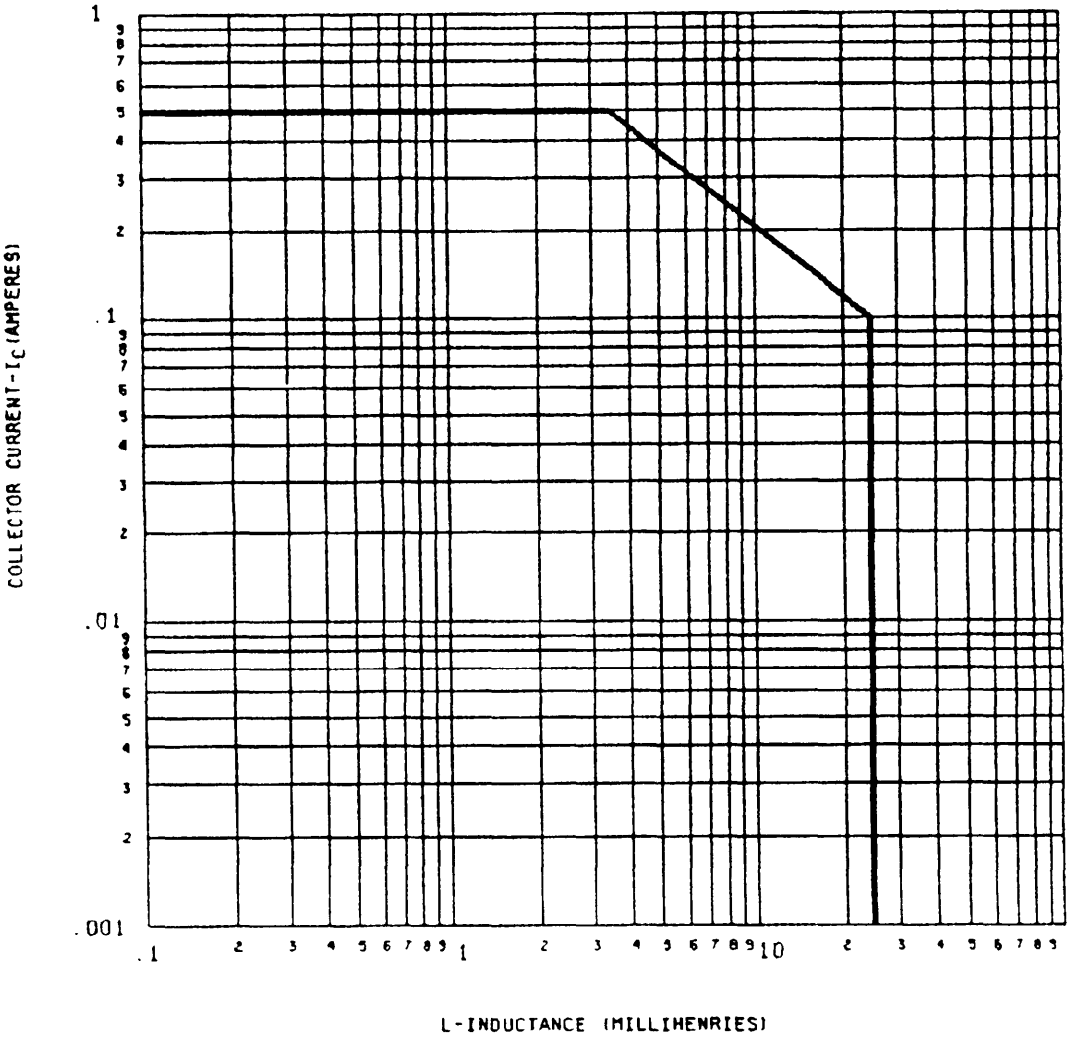
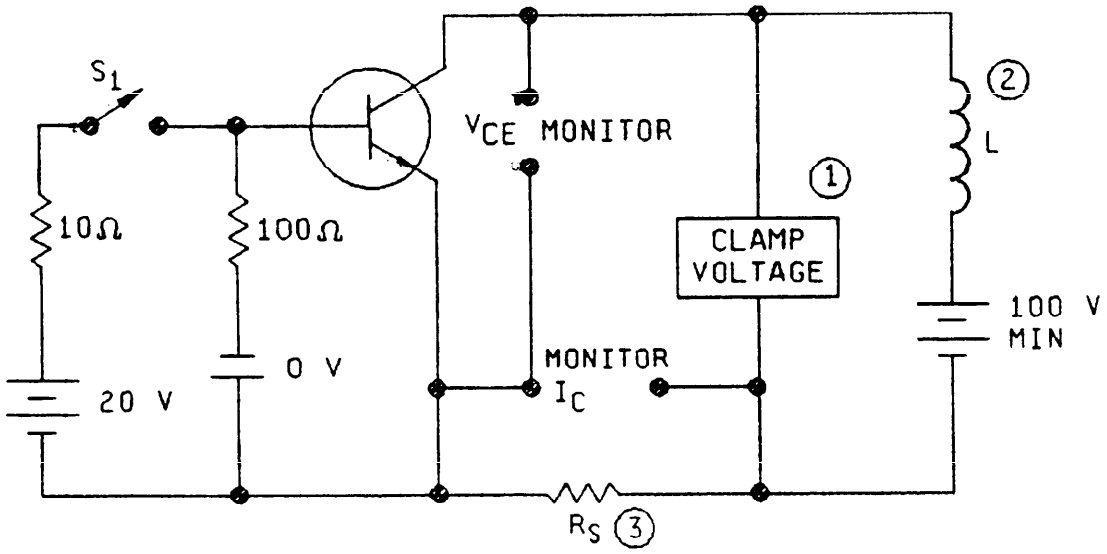


FIGURE 3. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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NOTES:
1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 25 mH at 1 A with a maximum dc resistance of 1 ohm. For reference only: Triad C-48u (center-tapped), or equivalent (see 4.4.5)
3. $R_S \leq 1$ ohm, 12 W, 1% tolerance max., (noninductive).

Procedure:
1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if clamp voltage not reached and maintained until the current returns to zero.
3. Perform specified endpoint tests.

FIGURE 4. Clamped inductive sweep test circuit.

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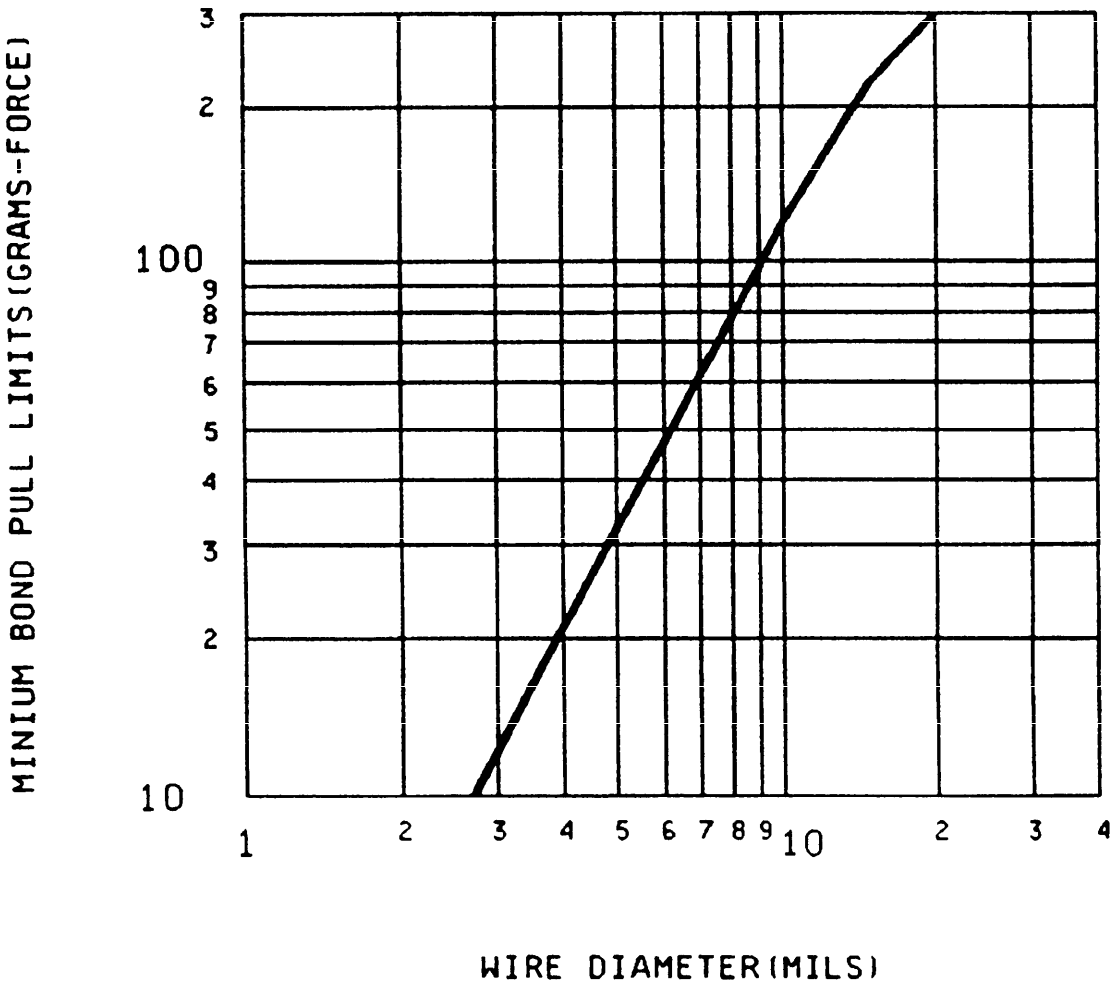


FIGURE 5 Minimum destructive bond-pull breaking-force vs. wire diameter for aluminum wire bonds.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level and type designation.

6.3 Changes from previous issue. Marginal notationss are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19, 85
DLA - ES
NASA - NA

User activities:
Army - AV, SM
Navy - AS, CG, MC, SH
Air Force - 13, 15

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1417)

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